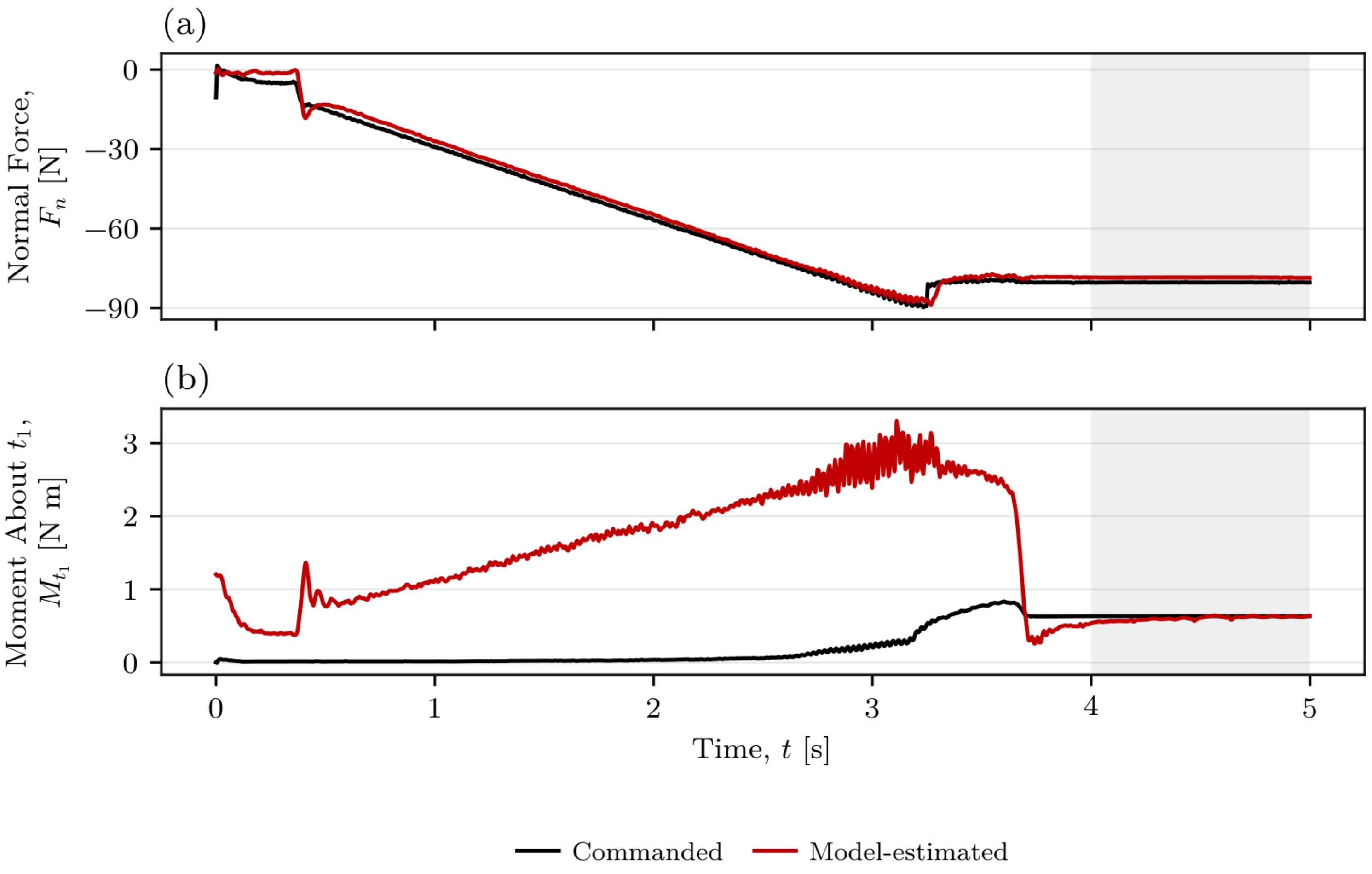
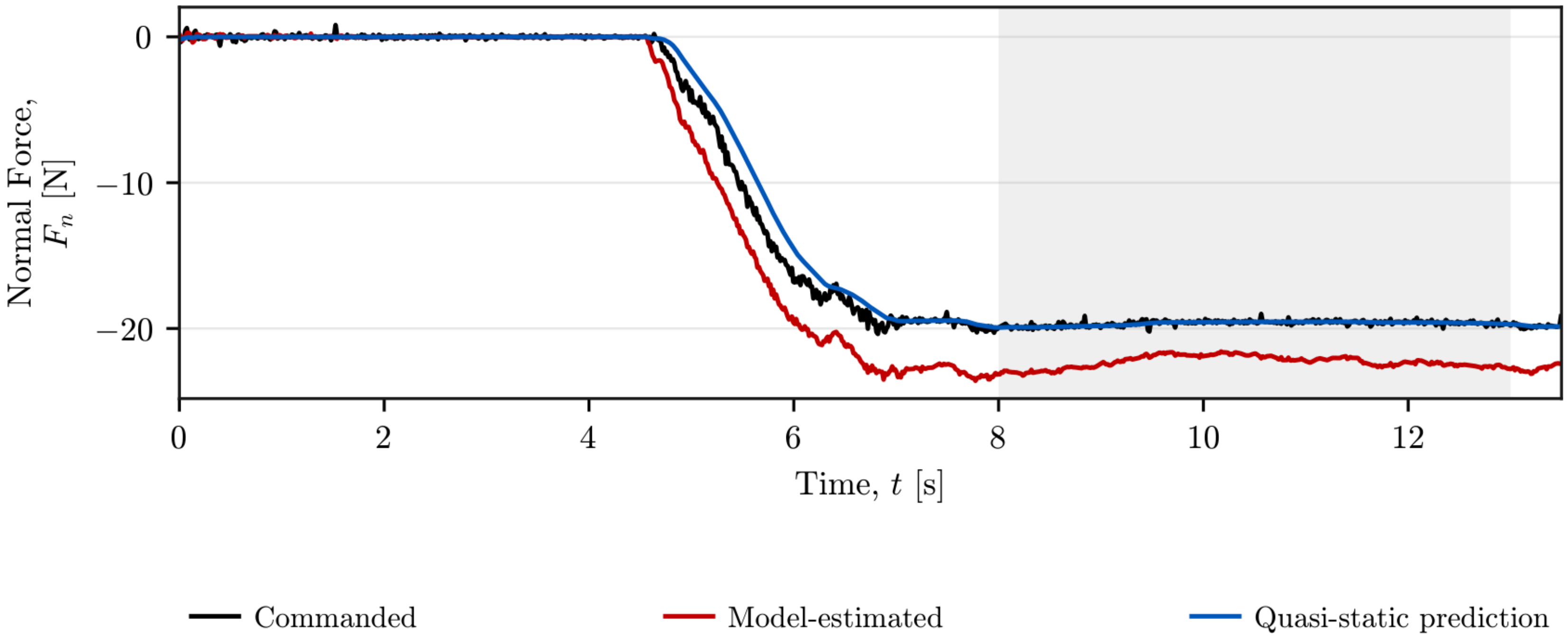
Commanded and estimated wrench



- Stationary force difference: 2.32%
- Stationary moment difference: 4.87%
- Supplementary plausibility evidence

Normal-force plausibility assessment

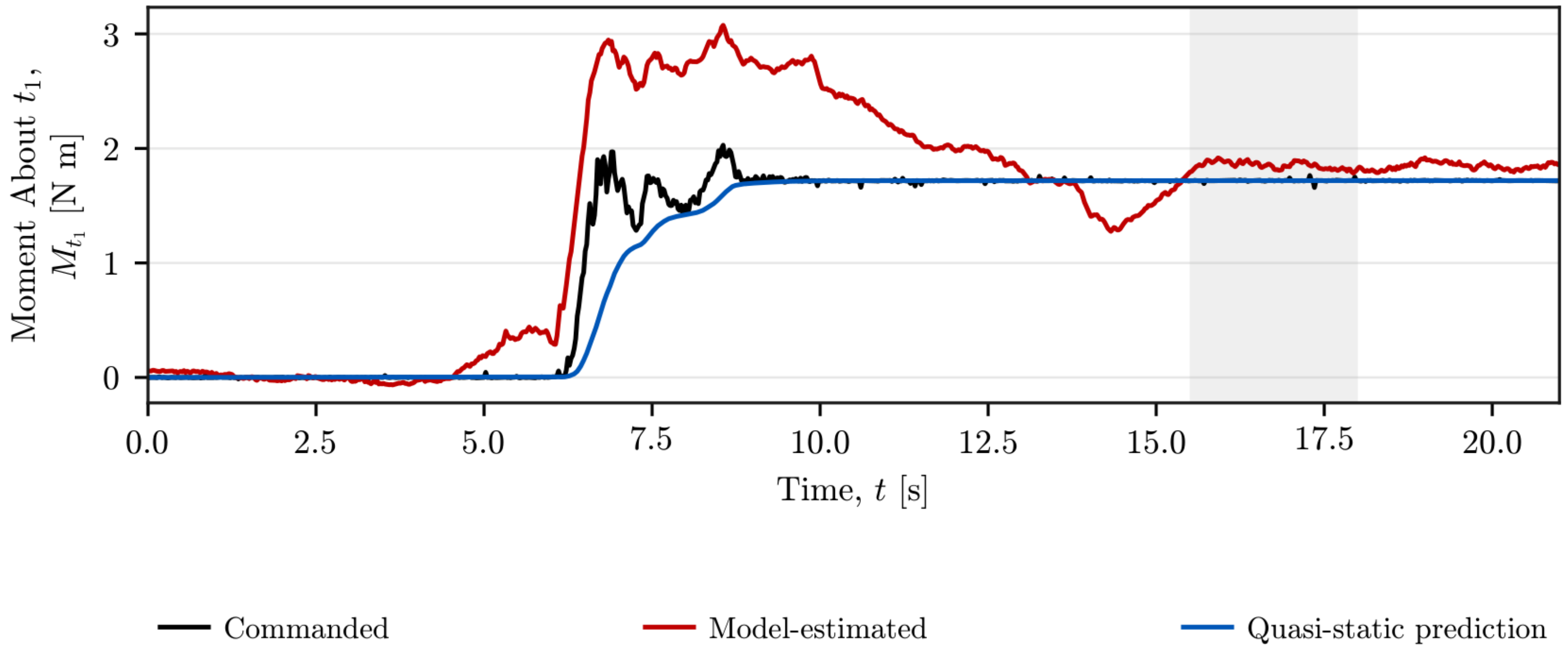
$$F_{n,\mathrm{qs}} = K_{p,n}\, n_s^{\top} (e_p - e_{p,0})$$



- Spring vs command: 0.05% difference
- Estimate vs command: 13.50% difference

Moment plausibility assessment

$$M_{t_1,\mathrm{qs}} = K_{R,t_1}\, t_1^\top (e_R - e_{R,0})$$



- Spring and commanded mean: 1.719 N m
- Model-estimated mean: 1.852 N m

End-effector pose representation

The EE frame is attached to the controlled tool centre point (TCP).

Position

$$p_{\mathrm{EE}} \in \mathbb{R}^3$$

3 translational degrees of freedom
TCP position in the robot base frame

Orientation

$$R_{\mathrm{EE}} \in SO(3)$$

3 rotational degrees of freedom
EE rotation relative to the robot base

Measured pose

$$T_{\mathrm{EE}} = \begin{bmatrix} R_{\mathrm{EE}} & p_{\mathrm{EE}} \\ 0^\top & 1 \end{bmatrix}$$

Obtained directly from the robot state for position and orientation feedback.

Null-space kinematics

7 joint coordinates control a 6-DOF tool pose: 3 position + 3 orientation.

$$\dim(\ker J) = 7 - \operatorname{rank}(J) = 7 - 6 = 1$$

At a nonsingular configuration: one independent redundant motion direction.

This freedom is a coordinated motion of the joints, not one designated joint.

$$J(q)\,\dot{q}_{\text{null}} = 0, \qquad J \in \mathbb{R}^{6\times 7}$$

J maps joint velocity to tool velocity: null-space motion gives zero tool velocity.

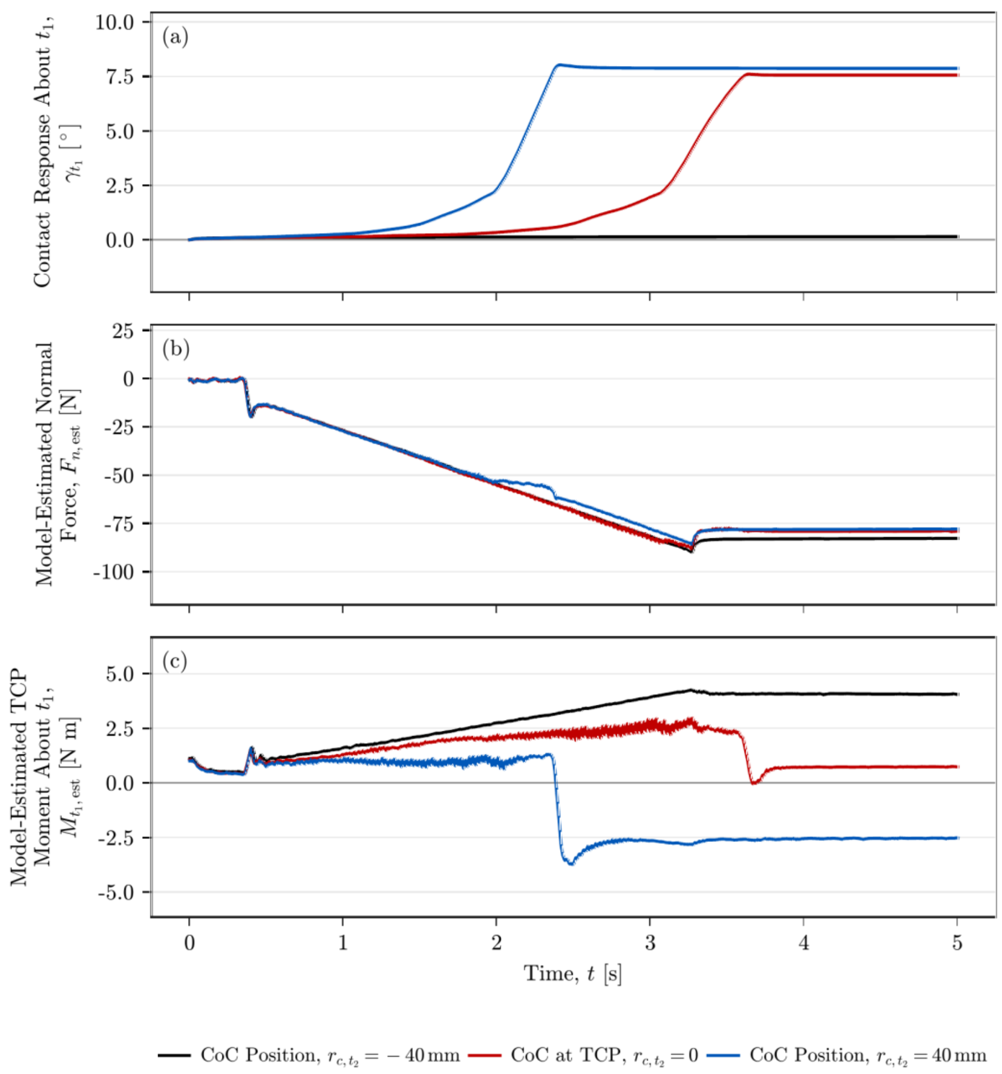
Two torque components control this internal motion:

$$\tau_{\text{null}} = \tau_d + \tau_\sigma$$

Damping torque: counters null-space joint motion.

Conditioning torque: steers towards a configuration with better motion capability.

Contact response, normal force and moment



Representative traces
Positive entry tilt

Similar final normal force

Moment reverses with CoC displacement

Force and moment: model-based robot estimates